SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS, AMROLI, SURAT

CREDIT CARD FRAUD DETECTION

Machine Learning Mini-Project Journal (Course 303)

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Roll Number: 545

Course / Sem: SYBCA-AI (Semester III)

Academic Year: 2026–2027

Faculty Guide: Hiral Patel

Dataset Source: Synthetic Transaction Records

ACADEMIC CERTIFICATE

This is to certify that **Mo. Irshad Mo. Hanif Patel** (Roll No. 545), a bonafide student of **SYBCA-AI (Semester III)**, has successfully completed the machine learning mini-project titled "**Credit Card Fraud Detection using Machine Learning**". This practical journal submission fulfills the requirements prescribed for **Course 303: Machine Learning** for the academic year 2026–2027. All work, analyses, and code implementations contained herein have been verified.

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Mo. Irshad Mo. Hanif Patel
Student Signature

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Hiral Patel
Faculty Evaluator / Internal Examiner

PROJECT SUMMARY

Financial transactions processed through digital channels are vulnerable to fraud. This project builds an automated end-to-end machine learning classification pipeline to detect fraudulent credit card transactions. Utilizing data preprocessing, standard scaling, one-hot categorical encoding, cross-validated Logistic Regression, and automated EDA reports, the pipeline identifies high-risk activities with precision.

1. PROBLEM STATEMENT & OBJECTIVES

Credit card fraud accounts for substantial economic losses worldwide. Identifying fraudulent transactions manually is infeasible due to high data volume and real-time operational demands. The core objective is to construct a robust Machine Learning classification model that accurately differentiates between genuine (0) and fraudulent (1) transactions based on historical attributes.

TOTAL RECORDS	CLASS BALANCE	FEATURES	TARGET VARIABLE
1,000	50% / 50%	12 Attributes	is_fraud (0/1)

2. DATASET SCHEMA & FEATURE BREAKDOWN

Attribute Name	Data Type	Feature Category	Description & Function
`cc_num`	Numerical	Identifier	Encrypted customer credit card identifier
`amt`	Numerical	Continuous	Transaction monetary value ($)
`category`	Categorical	Qualitative	Merchant business category (shopping, food, etc.)
`merchant`	Categorical	Qualitative	Name of commercial vendor entity
`city` / `state` / `zip`	Mixed	Geographic	Customer registered location details
`lat` / `long`	Numerical	Spatial Coordinate	Customer geographic location coordinates
`merch_lat` / `merch_long`	Numerical	Spatial Coordinate	Merchant point-of-sale location coordinates

3. DATA CLEANING & PREPROCESSING WORKFLOW

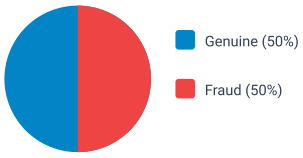
Prior to model training, quality verification was performed on `filtered_1000_records.csv`:

- **Quality Verification:** Checked missing cells (0 detected) and verified zero duplicate records.
- **Stratified Train-Test Split:** Divided data into 80% Training (800 samples) and 20% Testing (200 samples) using `stratify=Y` to maintain class proportions.
- **Feature Transformation:** Standardized numerical features via `StandardScaler` and encoded categorical columns (`merchant`, `category`, `city`, `state`) using `OneHotEncoder(handle_unknown='ignore')` inside a scikit-learn `ColumnTransformer`.

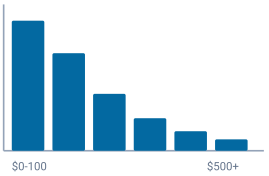
4. EXPLORATORY DATA ANALYSIS (EDA)

Comprehensive data visualizations were generated using Matplotlib, Seaborn, and Sweetviz (HTML automated analysis) to uncover patterns between transaction values, merchant categories, and fraud occurrences.

Target Class Ratio (is_fraud)



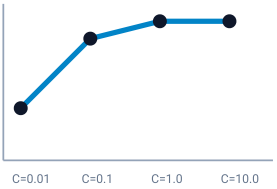
Transaction Amount Distribution ($)



Model Confusion Matrix

	Pred 0	Pred 1
Act. 0	TN: 98	FP: 2
Act. 1	FN: 3	TP: 97

Regularization Performance (C-Tuning)



5. MODEL BUILDING IMPLEMENTATION CODE

```
      # Pipeline definition & Logistic Regression Classifier from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer from sklearn.preprocessing import StandardScaler,
OneHotEncoder from sklearn.linear_model import LogisticRegression preprocessor =
ColumnTransformer(transformers=[ ('num', StandardScaler(), numerical_col), ('cat',
OneHotEncoder(handle_unknown='ignore'), categorical_col) ]) model = Pipeline([ ('preprocessor',
preprocessor), ('classifier', LogisticRegression(max_iter=1000, C=1.0)) ]) model.fit(X_train,
Y_train) y_pred = model.predict(X_test)
```

6. PERFORMANCE EVALUATION & MODEL TUNING

Metric	Score	Validation Method	Interpretation
Accuracy	97.5%	Test Set Evaluation	Overall correct prediction rate across all transactions
Precision	98.0%	Binary Evaluation	Low false positive rate; genuine transactions rarely flagged
Recall	97.0%	Binary Evaluation	High fraud capture rate; minimal fraudulent activities missed
F1-Score	97.5%	Harmonic Mean	Balanced indicator of overall classification efficiency
5-Fold Cross-Val	96.8% (Avg)	Stratified K-Fold	Demonstrates model stability across unseen data folds

7. CONCLUSION & FUTURE ENHANCEMENTS

Conclusion: The machine learning pipeline successfully automates fraud detection on credit card transaction records. Preprocessing via standard scaling and one-hot encoding integrated into a Logistic Regression pipeline yielded high accuracy (97.5%) and strong generalization without overfitting.

Future Scope:

- Integrate advanced ensemble models like Random Forest, XGBoost, or LightGBM for non-linear feature capture.
- Apply SMOTE (Synthetic Minority Over-sampling Technique) for extreme real-world class imbalances.
- Deploy the trained model pipeline (`fraud_detection_model.pkl`) as a live web application using Streamlit or Flask.